Australian Journal of Earth Sciences (1998) **45**, 673–675

Precise conventional and SHRIMP baddeleyite U–Pb age for the Binneringie Dyke, near Narrogin, Western Australia

A. A. NEMCHIN[1] AND R. T. PIDGEON[2]

[1]*Tectonics Special Research Centre, School of Applied Geology, Curtin University of Technology, PO Box U1987, Perth, WA 6845, Australia.*
[2]*School of Applied Geology, Curtin University of Technology, PO Box U1987, Perth, WA 6845, Australia.*

A precise baddeleyite U–Pb age of 2418 ± 3 Ma is reported for the westerly extension of the Binneringie Dyke in the south-western Yilgarn Craton of Western Australia. The Binneringie Dyke is a member of the large and extensive Widgiemooltha dyke swarm that trends east–west across the craton. This age is similar to ages of major dyke swarms in other Archaean Cratons and supports the hypothesis that dykes of the Widgiemooltha swarm are part of a worldwide Palaeoproterozoic mafic magmatic event at *ca* 2420 Ma.

Key words: baddeleyite, dolerite, dyke, Palaeoproterozoic, uranium–thorium–lead dating, Yilgarn Craton.

INTRODUCTION

Post-cratonisation mafic dykes in the Yilgarn Craton of Western Australia occur as large, widely spaced, gabbroic dykes that trend east–west across the craton and are referred to as the Widgiemooltha dyke swarm (Figure 1) (Soufoulis 1966; Hallberg 1987; Myers 1990). Other younger and denser swarms of relatively thin dykes are concentrated around the western and southern margins of the craton. There is little information available on the ages of these dykes and the purpose of this contribution is to report a precise baddeleyite U–Pb age for the large, east–west mafic dyke near Narrogin in southwestern Australia. This dyke is an extension of the Binneringie Dyke fracture system (Myers 1990), which extends over 550 km east–west across the Yilgarn Craton and is part of the Widgiemooltha dyke swarm.

RESULTS

The present sample (w430b) was taken from a coarse granophyric part of the dyke exposed in the Airport quarry near Narrogin (32°57′S, 117°07′E). At this locality the dyke is over 200 m wide.

About 40 baddeleyite plates between 50 and 100 μm in length were recovered from the 4 kg sample of granophyre, following the procedure described by Heaman and Machado (1992). U–Pb isotopic analyses of baddeleyite grains were made using conventional and SHRIMP techniques. Data are reported in tables and concordia plots with 1 σ errors. However, all ages quoted in this paper are reported with 95% confidence intervals. Conventional U–Pb analyses were made on three single plates of baddeleyite. Chemical separation of U and Pb was made using ion exchange techniques similar to those described by Nemchin *et al.* (1994) for zircon analyses, and uranium and lead were isotopically analysed on the Curtin University VG354 mass spectrometer. Results are presented in Table 1 and shown on a concordia plot (Figure 2).

U–Pb isotopic analyses of baddeleyite were also made on the Western Australian Consortium SHRIMP II at Curtin University. About 20 baddeleyite plates were mounted in epoxy resin together with chips of a Palabora baddeleyite standard. This has a conventional U–Pb age of 2059.8 ± 0.8 Ma (Heaman & Le Cheminant 1993). The analytical procedure for the baddeleyite was similar to that employed for zircon (Nelson 1997). Results of the SHRIMP analyses are also reported in Table 1 and data points are plotted on the concordia plot in Figure 2. Uranium concentrations, determined by SHRIMP analyses of 30 μm diameter areas within the baddeleyite grains, range from 1483 to 6013 ppm and suggest inhomogeneity between grains. However, this variability is almost certainly an artefact due to the orientation of the baddeleyite grains in the mount as discussed below. The more restricted range of uranium concentrations between grains, of 1239–1478 ppm, determined by isotope dilution (Table 1), is believed to reflect the true variation. The conventional data points are slightly discordant but fall on a discordia which has intersections with concordia at 2418 ± 3 Ma and 830 ± 105 Ma. The SHRIMP data show a strong reverse discordance but have consistent $^{207}Pb/^{206}Pb$ ratios (Figure 2) and the weighted average $^{207}Pb/^{206}Pb$ age determined from six individual SHRIMP spots is 2420 ± 7 Ma.

The possibility that the strong reverse discordance of the SHRIMP data points represents a real disturbance is doubtful as there is no evidence for this in the conventional dataset. It is more likely that the reverse discordance is due to crystal orientation effects as suggested by Wingate (1997) who showed that U and Pb sputtering and the U/Pb ratio determined during SHRIMP analyses of baddeleyite is dependent on the orientation of the crystals in the SHRIMP mount. The reverse discordance apparent in the five baddeleyite SHRIMP analyses (Figure 2) is therefore attributed to this orientation effect. However, this does not affect the $^{207}Pb/^{206}Pb$ age and the weighted mean $^{207}Pb/^{206}Pb$ age of 2420 ± 7 Ma is interpreted as a reliable estimate of the age of crystallisation of the dyke. This is identical, within error, to the age of 2418 ± 3 Ma determined from the

Copyright ©2001. All Rights Reserved.

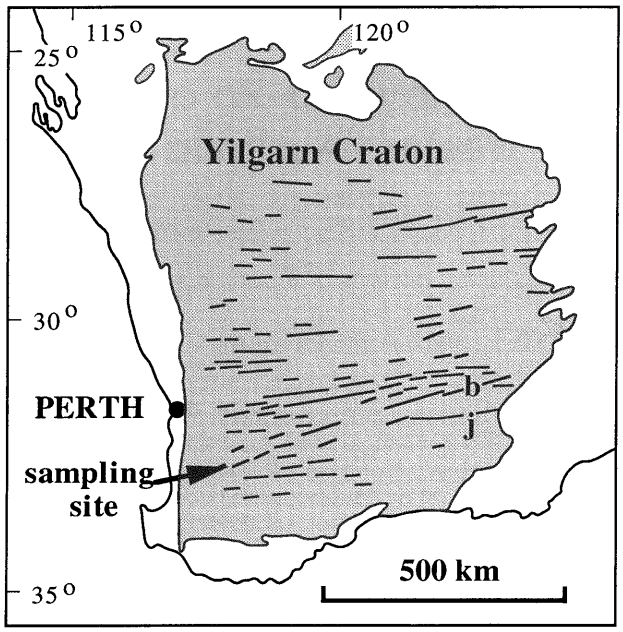


Figure 1 Distribution of the mafic dykes of the Widgiemooltha dyke swarm. b, Binneringie Dyke; j, Jimberlana Dyke.

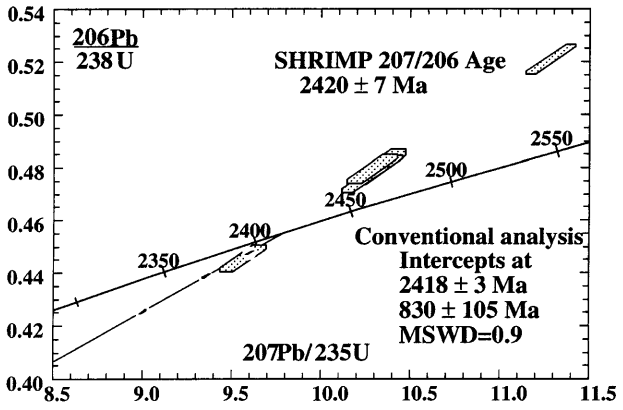


Figure 2 Concordia diagram showing SHRIMP and conventional results for baddeleyite from the granophyre sample.

upper intersection with concordia of a best-fit line through the three, slightly discordant, conventional data points (Figure 2, Table 1). As the conventional data points have the lower individual errors we have taken the upper intercept age of 2418 ± 3 Ma as the best estimate of crystallisation age of the dyke.

DISCUSSION

Rb–Sr results on a combination of whole-rock and mineral separates from the large east–west trending Jimberlana and parallel Celebration Dykes (Turek 1966; referred to by Fletcher *et al.* 1987) (which are also members of the Widgiemooltha dyke swarm) located in the Eastern Goldfields some 500 km northeast of the present sample, give a seven point isochron age of 2411 ± 52 Ma ($\lambda^{87}Rb = 1.47 \times 10^{-11}$ y^{-1}). Fletcher *et al.* (1987) reported an identical mineral and whole-rock Sm–Nd isochron age of 2411 ± 55 Ma (95% confidence limit) for these dykes. It has been shown by Heaman and Tarney (1989) and Lanyon *et al.* (1993), that Rb–Sr data can give results slightly younger as well as significantly older than the true age of dyke emplacement, and Sm–Nd results can also be influenced by problems of mixing of unrelated samples. However, this does not appear to be the case for the Rb–Sr and Sm–Nd ages of the Jimberlana Dyke, as these are the same within the uncertainties as the baddeleyite U–Pb age of the Binneringie Dyke presented in this paper. The similarity of the ages suggests that dykes of the Widgiemooltha swarm are the same age, now more precisely defined as 2418 ± 3 Ma, although further precise analyses on other dykes of the swarm are needed to confirm this.

The emplacement of mafic dykes is commonly associated with continental breakup (for example Heaman *et al.* 1992; Kamo & Gower 1994; Kamo *et al.* 1994) and 2418 Ma dykes in the Yilgarn Craton can be interpreted as dating the separation of the Yilgarn Craton from a larger Archaean continent as suggested by Myers *et al.* (1997). It has been suggested that the Widgiemooltha dyke swarm is part of a global magmatic event (Heaman 1997) involving the 2.42 Ga Scourie dyke swarm in the Lewisian of Scotland (Heaman & Tarney 1989), the 2.45 Ga dykes and basalts in

Table 1 U–Pb analyses of badelleyite from granophyre sample w430b (extension of Binneringie Dyke).

Spot	U (ppm)	Th (ppm)	Th/U	Pb (ppm)	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{206}Pb$	$^{208}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	Age(Ma) ($^{207}Pb/^{206}Pb$)
SHRIMP analyses										
w430b-1	2241	77	0.03	1005	10 470	0.15592 ± 86	0.01045 ± 100	0.4448 ± 50	9.56 ± 13	2412 ± 9
w430b-3	1483	20	0.01	714	11 444	0.15613 ± 96	0.00368 ± 108	0.4803 ± 55	10.34 ± 14	2414 ± 10
w430b-5	3650	173	0.05	1916	46 232	0.15751 ± 54	0.01548 ± 45	0.5197 ± 58	11.28 ± 14	2429 ± 6
w430b-8	6013	493	0.08	2931	38 670	0.15678 ± 46	0.02685 ± 41	0.4782 ± 53	10.34 ± 12	2421 ± 5
w430b-9	2187	74	0.03	1043	23 546	0.15651 ± 74	0.00904 ± 58	0.4749 ± 54	10.25 ± 13	2418 ± 8
w430b-10	1872	64	0.03	901	62 266	0.15590 ± 90	0.01279 ± 104	0.4786 ± 55	10.29 ± 14	2412 ± 10
TIMS analyses[a]										
Grain 1	1478			662	5208	0.15579 ± 7	0.00706 ± 1	0.4469 ± 7	9.600 ± 15	2411 ± 1
Grain 2	1239			570	2427	0.15375 ± 10	0.09837 ± 1	0.4248 ± 8	9.006 ± 18	2388 ± 1
Grain 3	1330			586	2674	0.15489 ± 11	0.01589 ± 1	0.4382 ± 8	9.358 ± 20	2401 ± 1

[a] Thermal ionisation mass spectrometry

Copyright ©2001. All Rights Reserved.

the Karelian craton (Amelin *et al.* 1995; Puchtel *et al.* 1997), the 2.46 Ga Great Dyke in the Zimbabwe craton (Hamilton 1977), the 2.42 Ga high-Mg tholeiitic dykes in the Vestfold craton, Antarctica (Collerson & Sheraton 1986) and the 2.42 Ga Bangalore dyke swarm in the Dharwar Craton (Ikrammuddin & Stueber 1976). Heaman (1997) suggested that the proliferation of flood basalt and associated magmatism near the Archaean–Proterozoic boundary may reflect a fundamental change in heat flux at the core–mantle boundary that heralded the breakup of a Late Archaean supercontinent. No temporally equivalent flood basalts or layered complexes have been identified in the Yilgarn Craton or in the Pilbara Craton to the north, although these may have been lost through subsequent uplift and erosion. On presently available geochronological results the Woongarra Volcanics, situated near the top of the Hamersley Group and dated at 2439 ± 10 Ma (Pidgeon & Horwitz 1991), are closest in time to the 2418 ± 3 Ma Widgiemooltha dykes.

Evidence that the 2.42 Ga dykes reflect major changes in the mantle and the breakup of a Late Archaean supercontinent should be contained in the age pattern of the dykes. Whether the dykes are of identical age or show trends in age that have some spatial pattern needs to be determined. Baddeleyite U–Pb ages are possibly the only available technique capable of providing sufficiently precise age determinations to resolve the required small age differences. The present precise age of 2418 ± 3 Ma for the western extension of the Binneringie Dyke provides a basis for a systematic investigation of the baddeleyite U–Pb ages of the Widgiemooltha dyke swarm that is needed to constrain models of the geological significance of the dykes.

ACKNOWLEDGEMENTS

We are grateful to S. Bodorkos, S. A. Wilde and P. A. Cawood for discussion and comments on early versions of this manuscript. J. S. Myers and W. G. Libby are thanked for their constructive reviews of the paper. This is Tectonics Special Research Centre Publication No. 32.

REFERENCES

Amelin Y. V., Heaman L. M. & Semenov V. S. 1995 U–Pb geochronology of layered mafic intrusions in the eastern Baltic Shield: implications for the timing and duration of Paleoproterozoic continental rifting. *Precambrian Research* **75**, 31–46.

Collerson K. D. & Sheraton J. W. 1986 Age and geochemical characteristics of a mafic dike swarm in the Archaean Vestfold Block, Antarctica: inferences about Proterozoic dike emplacement in Gondwanaland. *Journal of Petrology* **27**, 853–886.

Fletcher I. R., Libby W. G. & Rosman K. J. R. 1987 Sm–Nd dating of the 2411 Ma Jimberlarna dyke, Yilgarn Block, Western Australia. *Australian Journal of Earth Sciences* **34**, 523–525.

Hallberg J. A. 1987 Postcratonization mafic and ultramafic dykes of the Yilgarn Block. *Australian Journal of Earth Sciences* **34**, 135–149.

Hamilton J. 1977 Sr isotope and trace element studies on the Great Dyke and Bushveld mafic phase and their relation to Early Proterozoic magma genesis in Southern Africa. *Journal of Petrology* **18**, 24–52.

Heaman L. M. 1997 Global mafic magmatism at 2.45 Ga: remnants of an ancient large igneous province? *Geology* **25**, 299–302.

Heaman L. M. & L. E. Cheminant A. N. 1993 Paragenesis and U–Pb systematics of baddeleyite (ZrO_2). *Chemical Geology* **110**, 95–126.

Heaman L. M., Le Cheminant A. N. & Rainbird R. H. 1992 Nature and timing of Franklin igneous events, Canada: implications for a Late Proterozoic mantle plume and the break-up of Laurentia. *Earth and Planetary Science Letters* **109**, 117–131.

Heaman L. M. & Machado N. 1992 Timing and origin of midcontinental rift alkaline magmatism, North America: evidence from the Coldwell Complex. *Contributions to Mineralogy and Petrology* **110**, 289–303.

Heaman L. M. & Tarney J. 1989 U–Pb baddeleyite ages for the Scourie dike swarm, Scotland: evidence for two distinct intrusion events. *Nature* **340**, 705–708.

Ikrammuddin N. & Stueber A. M. 1976 Rb–Sr ages of Precambrian dolerites and alkaline dikes—Southern Mysore State, India. *Lithos* **9**, 235–241.

Kamo S. L. & Gower C. H. F. 1994 Note: U–Pb baddeleyite dating clarifies age of characteristic paleomagnetic remanence of Long Range dykes, southeastern Labrador. *Atlantic Geology* **30**, 159–262.

Kamo S. L., Krogh T. E. & Kamarapeli P. S. 1995 Age of the Grenville dyke swarm, Ontario–Quebec: implications for the timing of Iapetan rifting. *Canadian Journal Earth Sciences* **32**, 273–280.

Lanyon R., Black L. P. & Seitz H. M. 1993 U–Pb zircon dating of mafic dykes and its application to the Proterozoic geological history of the Vestfold Hills, East Antarctica. *Contributions to Mineralogy and Petrology* **115**, 184–203.

Myers J. S. 1990 Mafic Dyke swarms. *In: Geology and Mineral Resources of Western Australia*, pp. 126–127. Geological Survey of Western Australia Memoir **3**.

Myers J. S., Shaw R. D. & Tyler I. M. 1997 Tectonic evolution of Proterozoic Australia. *Tectonics* **15**, 1431–1446.

Nelson D. R. 1997 Compilation of SHRIMP U–Pb zircon geochronology data, 1996. *Geological Survey of Western Australia Record* **1997/2**.

Nemchin A. A., Pidgeon R. T. & Wilde S. A. 1994 Timing of Late Archaean granulite facies metamorphism in the southwestern Yilgarn Craton of Western Australia: evidence from U–Pb ages of zircons from mafic granulites. *Precambrian Research* **68**, 307–321.

Pidgeon R. T. & Horwitz R. C. 1991 The origin of olistoliths in Proterozoic rocks of the Ashburton Trough, Western Australia, using U–Pb isotopic characteristics. *Australian Journal of Earth Sciences* **38**, 55–63.

Puchtel I. S., Haase K. M. & Hofmann A. W. *et al.* 1997 Petrology and geochemistry of crustally contaminated komatiitic basalts from the Vetreny Belt, southeastern Baltic Shield: evidence for an early Proterozoic mantle plume beneath rifted Archean continental lithosphere. *Geochimica et Cosmochimica Acta* **61**, 1205–1222.

Soufoulis J. 1966 Widgiemooltha, WA. *Geological Survey of Western Australia 1:250 000 Geological Series Explanatory Notes.*

Turek A. 1966 Rubidium–strontium isotopic studies in the Kalgoorlie–Norseman area, Western Australia. PhD thesis. Australian National University, Canberra (unpubl.).

Wingate M. T. D. 1997 Testing Precambrian continental reconstructions using ion microprobe U–Pb baddeleyite geochronology and palaeomagnetism of mafic igneous rocks. PhD thesis. Australian National University, Canberra (unpubl.).

Received 23 March 1998; accepted 22 April 1998

Copyright ©2001. All Rights Reserved.